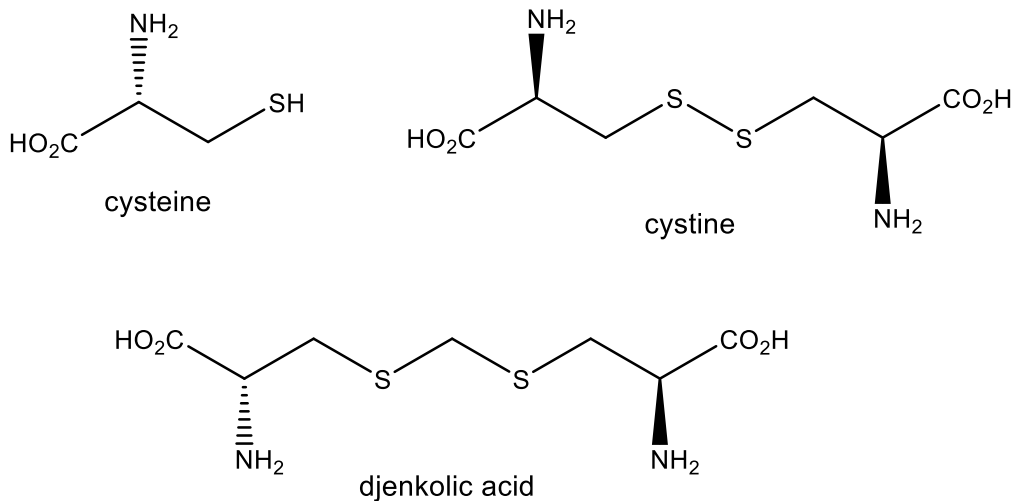


- 22** Cysteine and cystine are important biochemicals in which the former is biosynthesised from the latter. Cysteine is a naturally occurring α -amino acid, while djenkolic acid is a non-protein amino acid found in the djenkol beans of the Southeast Asian plant.

The three pK_a values of cysteine are 1.71, 8.33 (belonging to the thiol, $-SH$), and 10.78.



Which statement about cysteine, cystine and djenkolic acid is true?

- A** Cystine is a *meso* compound, hence it is non-chiral.
- B** Cysteine exists solely in its zwitterionic form at pH 5.02.
- C** Cystine and cysteine share the same empirical formula.
- D** There are four stereoisomeric forms of djenkolic acid because there are two chiral atoms in djenkolic acid.

23 Isoleucine is a naturally occurring compound found in fruit juices, vegetables, and other